

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, II: THE TRACE OBSTRUCTION AND THE K-HOMOLOGICAL BULK

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ABSTRACT. Paper I isolated a virtual KK -class $x_p = T_p - qI \in KK^0(\mathbb{C}^n, \mathbb{C}^n) = M_n(\mathbb{Z})$, arising from the Hecke–Laplacian $\Delta_p = (q+1)I - T_p = 1 - x_p$ of a finite $(q+1)$ -regular graph Y , and left open whether x_p could be realized by a genuine, non-negative C^* -correspondence, so that Δ_p ’s cokernel $\mathbb{Z} \oplus \text{Jac}(Y)$ — the critical group of Y , whose order is a product of Frobenius traces $\prod(p+1 - a_p(f))$ — would be the K_0 -group of an honest Cuntz–Pimsner algebra. This paper proves that no such realization exists, by an elementary and then a deeper obstruction, and identifies where the arithmetic must live instead.

The companion linearization $C = \begin{pmatrix} T_p & -qI \\ I & 0 \end{pmatrix}$, exactly row/column-equivalent to Δ_p over $\mathbb{Z}$, has $\text{Tr}(C) = 0$, $\text{Tr}(C^2) = -n(q-1)$, strictly negative for every $q \geq 2$ and every $(q+1)$ -regular graph. Since $\text{Tr}(B^2) \geq 0$ for any non-negative matrix B , and trace is a similarity invariant, **no matrix of the same size and the same spectrum as C can be non-negative** (Theorem 1.2) — an unconditional, four-line fact. Every dilation to a *larger* non-negative matrix, of the kind an auxiliary-vertex or Drinen–Tomforde construction would attempt, is ruled out as well, and unconditionally: shift equivalence preserves the nonzero spectrum with multiplicities — proved here by an elementary two-way injection of generalized eigenspaces (Lemma 1.4) — hence preserves $\text{Tr}(\cdot^2)$, closing the realization problem of Paper I, Remark 3.11, in the negative for matrices of every size (Theorem 1.5).

We then show that no repair by $\mathbb{Z}/2$ -grading is available: a formal difference of two positive correspondences is not itself a correspondence, and no Cuntz–Pimsner-type functor converts a virtual difference into a single algebra realizing that difference’s index (Section 2); if one existed, it would trivialize the realization problem for every virtual K -theory class whatsoever, which is itself evidence against its existence.

The arithmetic is instead formalized where no positivity is required: the Hecke–Dirac operator D_p defines an honest Fredholm module over $C(V_Y)$ with $\text{ind}(D_p) = \mathbb{Z} \oplus \text{Jac}(Y)$ exactly (Theorem 3.3), inside $KK^0(C(V_Y), C(V_Y)) \cong M_n(\mathbb{Z})$. We record, rather than disguise, that for a single finite graph this KK -group is nothing but a matrix ring and carries no content beyond ordinary linear algebra; the paper closes by locating the correct venue for a genuinely non-commutative sequel in the projective limit over a congruence tower of graphs, where the coefficient algebra becomes an infinite, totally disconnected profinite space and KK^0 ceases to be finite-dimensional. As a first exact instance of the tower philosophy, we compute four abelian $\mathbb{Z}/\ell^k$ voltage towers over K_4 to nine levels: the torsion obeys the Iwasawa law $\text{ord}_\ell |\text{Jac}(Y_k)| = \mu\ell^k + \lambda k + \nu$ exactly at every computed level, with $(\mu, \lambda) = (3, 1), (0, 3), (2, 3), (0, 1)$, positive μ occurring freely, and the Smith normal forms exhibiting λ as the number of persistently growing cyclic factors. The canonical combinatorial model of the congruence tower itself — the Iwahori path tower — obeys instead an exact pseudo-Iwasawa law with $\lambda = -1 < 0$, impossible for a $\mathbb{Z}_p[[T]]$ -characteristic ideal: the non-normal tower is not an Iwasawa module, and the conjecture for it is restated over the Iwahori–Hecke algebra, with its naive commutative form withdrawn.

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Paper II of the series begun in *Algorithmic non-commutative class field theory, I: the Langlands–thermodynamic bridge over Bruhat–Tits trees*. This paper answers, negatively, the realization question left open there as Remark 3.11 (equivalently Remark 3.6 in some section numberings of the same manuscript): whether the companion linearization of the Ihara pencil admits a positive C^* -correspondence realizing it directly. Verification scripts are referenced inline; none require network access.

STATUS OF THE RESULTS, STATED IN ADVANCE

As in Paper I, each numbered result is labelled by how it is entitled to be believed. Theorem 1.2 is proved here in full, elementarily, using nothing beyond the fact that trace is a similarity invariant and that non-negative matrices have non-negative squared-trace. Theorem 1.5, which rules out the auxiliary-vertex and Drinen–Tomforde schemes of any size, is likewise unconditional: the fact it needs — that shift equivalence preserves the nonzero spectrum with multiplicities — is proved here as the elementary Lemma 1.4 by a two-way injection of generalized eigenspaces, and no external citation is load-bearing for it. Theorem 3.3 is proved in full and requires no citation beyond standard finite-dimensional linear algebra dressed in Fredholm-module language. Section 3 is exploratory: it states a research programme, not a theorem, and we say so throughout.

1. THE TRACE OBSTRUCTION

1.1. Setup, recalled from Paper I. Y is a finite connected $(q+1)$ -regular graph on n vertices, $T_p \in M_n(\mathbb{Z})$ its adjacency (Hecke) operator, $\Delta_p := (q+1)I_n - T_p$ its Hecke–Laplacian, and $\text{Jac}(Y)$ its critical group, with

$$\text{coker}_{\mathbb{Z}}(\Delta_p) \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad |\text{Jac}(Y)| = \kappa(Y) = \frac{1}{n} \prod_f (p+1 - a_p(f)) = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p),$$

the product over the weight-two newforms f whose Hecke eigenvalues occur in $\text{Spec } T_p$, by Kirchhoff’s theorem together with the Eichler–Shimura and Jacquet–Langlands correspondences. Introduce the companion linearization

$$C := \begin{pmatrix} T_p & -qI_n \\ I_n & 0 \end{pmatrix} \in M_{2n}(\mathbb{Z}),$$

of the Ihara pencil $1 - uT_p + qu^2$, so that $\det(I_{2n} - uC) = \det(I_n - uT_p + qu^2I_n)$. By Schur elimination against the invertible block I_n in the lower right of $I_{2n} - C = \begin{pmatrix} I_n - T_p & qI_n \\ -I_n & I_n \end{pmatrix}$, one has the exact integral equivalence

$$\begin{pmatrix} I_n & -qI_n \\ 0 & I_n \end{pmatrix} (I_{2n} - C) \begin{pmatrix} I_n & 0 \\ I_n & I_n \end{pmatrix} = \begin{pmatrix} \Delta_p & 0 \\ 0 & I_n \end{pmatrix}, \quad \text{hence} \quad \text{coker}_{\mathbb{Z}}(I_{2n} - C) \cong \mathbb{Z} \oplus \text{Jac}(Y),$$

with no denominators at any step, since both flanking matrices are unimodular. So C is not merely spectrally related to Δ_p ; it is honestly row/column-equivalent to it, and any non-negative realization of C solves the realization problem for Δ_p with no loss.

1.2. The elementary, unconditional obstruction.

Lemma 1.1. *For every $(q+1)$ -regular simple graph Y on n vertices, $\text{Tr}(C) = 0$ and*

$$\text{Tr}(C^2) = -n(q-1).$$

Proof. $\text{Tr}(C) = \text{Tr}(T_p) + \text{Tr}(0) = 0$ since Y is simple (no loops). In block form,

$$C^2 = \begin{pmatrix} T_p^2 - qI_n & -qT_p \\ T_p & -qI_n \end{pmatrix},$$

so $\text{Tr}(C^2) = \text{Tr}(T_p^2) - qn - qn = \text{Tr}(T_p^2) - 2qn$. Since T_p is symmetric with entries in $\{0, 1\}$, $\text{Tr}(T_p^2) = \sum_{i,j} (T_p)_{ij}^2 = 2|E_Y| = (q+1)n$ by regularity. Hence $\text{Tr}(C^2) = (q+1)n - 2qn = -n(q-1)$. $\square$

This has been verified directly on the eleven graphs of Table 1, in every case with exact agreement; the lemma is general and the table is confirmation, not evidence.

Theorem 1.2 (Unconditional obstruction: no equal-size, equal-spectrum realization). *Let $q \geq 2$. No matrix $\tilde{C} \in M_{2n}(\mathbb{Z}_{\geq 0})$ similar to C — in particular, no non-negative matrix with the same characteristic polynomial as C — exists.*

Proof. Trace is invariant under similarity, so $\text{Tr}(\tilde{C}) = \text{Tr}(C) = 0$ and $\text{Tr}(\tilde{C}^2) = \text{Tr}(C^2) = -n(q-1) < 0$ would be forced. But for any matrix $B \in M_N(\mathbb{Z}_{\geq 0})$, $\text{Tr}(B^2) = \sum_{i,j} B_{ij}B_{ji} \geq 0$, a sum of products of non-negative integers. Contradiction. $\square$

Remark 1.3. Theorem 1.2 needs nothing beyond Lemma 1.1 and the one-line non-negativity of $\text{Tr}(B^2)$ for non-negative B ; it requires no appeal to symbolic dynamics, and it already rules out any construction that keeps C 's size fixed at $2n$ — for instance, any attempt to directly reinterpret the entries of C itself, without adding vertices, as a graph adjacency matrix.

1.3. The obstruction to dilation of any size. The constructions actually attempted in the course of this project (auxiliary sink vertices, Drinen–Tomforde-style desingularization, cascading intermediate nodes) all enlarge the matrix, replacing C by some $\tilde{C} \in M_N(\mathbb{Z}_{\geq 0})$, $N > 2n$, related to C not by similarity but by *shift equivalence*: matrices $R \in M_{2n \times N}(\mathbb{Z})$, $S \in M_{N \times 2n}(\mathbb{Z})$ and a lag ℓ with

$$CR = R\tilde{C}, \quad SC = \tilde{C}S, \quad RS = C^\ell, \quad SR = \tilde{C}^\ell.$$

Shift equivalence is the correct and standard notion for comparing presentation matrices of possibly different sizes whose associated shifts of finite type (equivalently, Cuntz–Krieger algebras) are to be considered the same dynamical, and K -theoretic, object: the Bowen–Franks group $\text{coker}_{\mathbb{Z}}(I - \tilde{C})$ is a shift-equivalence invariant, which is exactly why enlarging the matrix could in principle have been a way to keep $\text{coker}(I - \tilde{C}) \cong \mathbb{Z} \oplus \text{Jac}(Y)$ while repairing non-negativity.

Lemma 1.4 (Shift equivalence preserves the nonzero spectrum). *Let $A \in M_a(\mathbb{C})$, $B \in M_b(\mathbb{C})$ be lag- ℓ shift equivalent: there exist $R \in M_{a \times b}$, $S \in M_{b \times a}$ with*

$$AR = RB, \quad SA = BS, \quad RS = A^\ell, \quad SR = B^\ell.$$

Then for every $\lambda \neq 0$ the generalized eigenspaces satisfy $\dim V_\lambda(A) = \dim V_\lambda(B)$; hence the nonzero spectra of A and B agree as multisets, and $\text{Tr}(A^k) = \text{Tr}(B^k)$ for every $k \geq 1$, including $k < \ell$.

Proof. Let $v \in V_\lambda(A) = \ker(A - \lambda)^{k_0}$, $\lambda \neq 0$. Then $(B - \lambda)^{k_0}Sv = S(A - \lambda)^{k_0}v = 0$, so S maps $V_\lambda(A)$ into $V_\lambda(B)$. If $Sv = 0$ then $A^\ell v = RSv = 0$; but A restricted to $V_\lambda(A)$ has sole eigenvalue $\lambda \neq 0$, hence is invertible there, forcing $v = 0$. So $S|_{V_\lambda(A)}$ is injective. Symmetrically, R maps $V_\lambda(B)$ injectively into $V_\lambda(A)$. Two finite-dimensional spaces injecting into each other have equal dimension. Algebraic multiplicity equals the dimension of the generalized eigenspace, and $\text{Tr}(M^k) = \sum_\lambda m_\lambda \lambda^k$ with zero eigenvalues contributing nothing for $k \geq 1$; the trace identity follows. $\square$

Theorem 1.5 (Unconditional obstruction: no dilation of any size). *Let $q \geq 2$. No matrix $\tilde{C} \in M_N(\mathbb{Z}_{\geq 0})$, of any size N , is shift equivalent to C . Consequently no auxiliary-vertex, desingularization, or other enlargement scheme produces a non-negative integer matrix presenting the same K -theoretic and dynamical data as C .*

Proof. By Lemma 1.4, shift equivalence would force $\text{Tr}(\tilde{C}^2) = \text{Tr}(C^2) = -n(q-1) < 0$, contradicting $\text{Tr}(\tilde{C}^2) = \sum_{i,j} \tilde{C}_{ij}\tilde{C}_{ji} \geq 0$. $\square$

Remark 1.6 (Relation to the spectral conjecture literature). An earlier draft stated Theorem 1.5 conditionally on the resolution of the Boyle–Handelman spectral conjecture by Kim–Ormes–Roush. That was an error of economy, not of substance: the deep content of that literature is the *converse* direction — which trace-positive spectra *are* realizable by non-negative matrices

— whereas the direction needed here, invariance of the nonzero spectrum under shift equivalence, is the elementary Lemma 1.4. The theorem is unconditional and self-contained; the spectral-conjecture literature remains relevant context for the converse question, and its citations remain unverified by us.

Remark 1.7 (What is, and is not, ruled out). Theorem 1.5 rules out any *faithful* dilation of C — one preserving its nonzero spectrum, hence its actual Ihara/Hecke dynamical content, which is the only kind of dilation that would let the resulting graph algebra be honestly called a realization of the Hecke correspondence. It does not, and cannot, rule out the bare abstract existence of *some* unrelated non-negative matrix D with $\text{coker}_{\mathbb{Z}}(I - D) \cong \mathbb{Z} \oplus \text{Jac}(Y)$: by Kirchberg–Phillips classification, a Kirchberg algebra realizing any prescribed pair (K_0, K_1) with K_0 finitely generated and K_1 free exists unconditionally, with no relation to T_p required. Such an algebra would answer no question of interest here; the content of this paper is that the *natural* one, built from C itself, does not exist.

2. THE VIRTUAL CORRESPONDENCE BARRIER

Given Theorems 1.2–1.5, it is natural to ask whether the obstruction can be sidestepped by weakening what counts as a “correspondence,” for instance by allowing a $\mathbb{Z}/2$ -graded or super-structure in which T_p is placed in even degree and qI_n in odd degree, with the hope that a graded Cuntz–Pimsner-type construction applied to $\mathcal{E}^+ \oplus \mathcal{E}^-$ (with $[\mathcal{E}^+] = T_p$, $[\mathcal{E}^-] = qI_n$, both honest positive correspondences) might produce a single algebra whose K_0 -group is the cokernel of $1 - ([\mathcal{E}^+] - [\mathcal{E}^-]) = \Delta_p$.

Proposition 2.1. *No such construction exists. More precisely: for any C^* -algebra A and any two genuine A – A correspondences $\mathcal{E}^+, \mathcal{E}^-$, the Cuntz–Pimsner algebra of the direct sum $\mathcal{E}^+ \oplus \mathcal{E}^-$ — graded or not — has K -theoretic index map $1 - [\mathcal{E}^+] - [\mathcal{E}^-]$, not $1 - [\mathcal{E}^+] + [\mathcal{E}^-]$; a $\mathbb{Z}/2$ -grading records parity for the purpose of assigning classes to KK^0 versus KK^1 , and does not introduce a sign into the index computation of a single Cuntz–Pimsner algebra built from an honestly positive module.*

Discussion, in place of a proof. The Cuntz–Pimsner construction is a functor from the category of genuine Hilbert A – A -bimodules (with a fixed left action by compacts, or by adjointable operators, as appropriate) to C^* -algebras, and its K -theory is computed, via Pimsner’s exact sequence, from the class $[\mathcal{E}]$ of the *single* bimodule fed to it, and the root of the constraint is the axiom $\langle \xi, \xi \rangle \geq 0$ of the Hilbert C^* -module inner product: every channel of the completed module carries non-negative norm, so no summand can act with a negative sign in the completion. There is no step in the construction at which a summand of the bimodule can be assigned a formal minus sign: the bimodule $\mathcal{E}^+ \oplus \mathcal{E}^-$ is a perfectly good, positive, correspondence in its own right, with class $[\mathcal{E}^+ \oplus \mathcal{E}^-] = [\mathcal{E}^+] + [\mathcal{E}^-] = T_p + qI_n$, and its Cuntz–Pimsner algebra has index map $1 - (T_p + qI_n)$, the *opposite* sign adjustment from the one wanted. Introducing a $\mathbb{Z}/2$ -grading changes which Kasparov group (KK^0 or KK^1) a resulting class is assigned to; it does not alter this computation. We are not aware of, and have not been able to construct, any variant Cuntz–Pimsner-type functor that takes a *formal difference* of two correspondences as input and returns an algebra realizing that difference as an index. Were such a functor to exist, it would apply to every virtual class in every $KK^0(A, A)$, converting every formal difference of finitely generated projective modules into a single algebra realizing it as an index map, and would in particular apply to the underlying integer matrix problem addressed by Theorems 1.2–1.5, trivializing it; the fact that those theorems hold, and are elementary, is itself evidence that no such functor exists. $\square$

Remark 2.2 (The obstruction is ontological, not technical). The negative block $-qI_n$ in C is not an artifact of a particular presentation that a cleverer bookkeeping device — grading, desingularization, or otherwise — could absorb. It is a faithful record of the non-backtracking correction

subtracted in the Ihara recursion $T \cdot T_1 = T_2 + (q+1)T_0$: the class $x_p = T_p - qI_n$ is the class of a correspondence *minus* the class of q copies of the identity correspondence, and Theorem 1.2 shows this difference is not, for any $q \geq 2$, again the class of a correspondence. A virtual class that is provably not effective cannot be made effective by re-describing it; it can only be relocated to a category that does not demand effectivity. That category is KK -theory, to which we now turn.

3. THE K-HOMOLOGICAL BULK, AND THE CONGRUENCE TOWER

3.1. The Hecke–Dirac operator and its index.

Definition 3.1 (Hecke–Dirac operator). Fix an orientation E_Y^+ of the edges of Y and let $\partial: \mathbb{Z}^{E_Y^+} \rightarrow \mathbb{Z}^{V_Y}$, $\partial(f) = t(f) - o(f)$, be the simplicial boundary map, so $\partial^t: \mathbb{Z}^{V_Y} \rightarrow \mathbb{Z}^{E_Y^+}$ is its transpose (coboundary). Define the self-adjoint operator

$$D_p := \begin{pmatrix} 0 & \partial \\ \partial^t & 0 \end{pmatrix} \quad \text{on} \quad \ell^2(V_Y) \oplus \ell^2(E_Y^+), \quad D_p^2 = \begin{pmatrix} \partial\partial^t & 0 \\ 0 & \partial^t\partial \end{pmatrix},$$

i.e. $D_p = d + d^*$ is the discrete Hodge–de Rham operator of the graph, with even part the vertex (Hodge 0-)Laplacian $\partial\partial^t = \Delta_p$ and odd part the edge (Hodge 1-)Laplacian $\partial^t\partial$.

Lemma 3.2. $\partial\partial^t = \Delta_p = (q+1)I - T_p$ on $\ell^2(V_Y)$.

Proof. $(\partial\partial^t)_{uv} = \sum_{f \in E_Y^+} \partial_f(u) \partial_f(v)$; the diagonal term is the degree $q+1$ (each incident edge contributes ± 1 squared) and the off-diagonal term at $u \neq v$ is -1 for each edge joining u, v (with a sign cancellation independent of orientation choice, since $\partial_f(u)\partial_f(v) = -1$ whenever f joins u, v regardless of which endpoint is called $o(f)$), giving exactly $(q+1)I - T_p$. $\square$

Theorem 3.3 (Fredholm module and index). $(C(V_Y), \ell^2(V_Y) \oplus \ell^2(E_Y^+), D_p)$ is an even Fredholm module over $A := C(V_Y)$, defining a class $[D_p] \in KK^0(A, A) \cong M_n(\mathbb{Z})$, $n = |V_Y|$, and

$$\begin{aligned} \text{ind}(D_p) &:= [\ker D_p] - [\text{coker } D_p] \in K_0(A), \quad \ker D_p \cong \mathbb{Z} \oplus \mathbb{Z}^{\text{rank } H_1(Y)}, \\ \text{coker}(D_p|_{\ell^2(V_Y)}) &\cong \mathbb{Z} \oplus \text{Jac}(Y), \end{aligned}$$

so that the even part of the index recovers Δ_p 's cokernel exactly: K_0 -theoretically, $[D_p^2|_{\ell^2(V_Y)}]$ has cokernel $\mathbb{Z} \oplus \text{Jac}(Y)$, with no positivity requirement anywhere in the construction — D_p is self-adjoint, not positive, and ∂, ∂^t are integer matrices of arbitrary sign.

Proof. D_p is a finite self-adjoint matrix, hence trivially Fredholm (bounded, and every finite-rank perturbation of 0 is compact), giving a class in $KK^0(A, A)$ since $A = C(V_Y) = \mathbb{C}^n$ acts diagonally on $\ell^2(V_Y)$ and trivially, via a chosen embedding of V_Y -labels, on $\ell^2(E_Y^+)$ compatible with ∂ 's A -bimodule structure. $\ker D_p = \ker \partial^t \oplus \ker \partial$; $\ker \partial^t$ is the space of harmonic 0-cochains, $\cong \mathbb{Z}$ (the constants, Y connected), and $\ker \partial \cong H_1(Y; \mathbb{Z}) \cong \mathbb{Z}^{\text{rank } H_1(Y)}$, the cycle space. The even part follows from Lemma 3.2 and Theorem 3.5(i) of Paper I. $\square$

Remark 3.4 (The honest scope of this theorem). This is a correct and complete restatement, in Fredholm-module language, of results already proved by direct linear algebra in Paper I. We record it because the language matters for what comes next: D_p 's self-adjointness, not positivity, is all that KK -theory demands, and Theorem 1.2 shows this is not a matter of convenience but of necessity — no positive substitute for $D_p^2 - qI$ -type data exists. It is not, however, a new theorem about Y ; every computation in it already appears in Paper I under other names.

3.2. Why nothing new happens for a single graph.

Remark 3.5 (Finite-dimensional triviality). For $A = C(V_Y) = \mathbb{C}^n$, $KK^0(A, A) \cong M_n(\mathbb{Z})$ as a ring under the Kasparov product, and the product of two classes represented by integer matrices is exactly their matrix product. Consequently, any identity one might hope to derive by taking a Kasparov product of $[D_p]$ with the Hecke correspondence class $[\mathcal{E}_p] = T_p$ is nothing more than the corresponding matrix identity, already available by direct computation. We tested the specific proposed identity $[D_p] \otimes [\mathcal{E}_p] \stackrel{?}{=} [D_{p^2}] + q[D_0]$ under both natural readings of $[D_p]$ (as Δ_p and as $T_p - qI$) against the matrices directly: neither matches. What is true, and is already Lemma 2.1 of Paper I, is the unadorned recursion $T_p \cdot T_p = T_2 + (q+1)I$, with no reference to D_p or to any Kasparov product required to state or prove it. ***KK-theory for a single finite graph adds no content beyond $M_n(\mathbb{Z})$*** ; dressing an identity in Kasparov-product language does not make it a new theorem when the underlying ring is already $M_n(\mathbb{Z})$ with ordinary matrix multiplication.

3.3. The congruence tower: where new content could appear. The finite-dimensionality identified in Remark 3.5 is specific to working at a single level Y . The genuine non-commutative geometry of the Langlands correspondence, if it is to be found at all along this route, should appear only in the limit over a *tower* of graphs.

Conjecture 3.6 (Congruence tower and the profinite KK -limit). Let $(Y_k)_{k \geq 0}$ be the tower of quotient graphs $Y_k = \Gamma_0(p^k) \backslash X$ for the Bruhat–Tits tree X of $\mathrm{PGL}_2(\mathbb{Q}_p)$, with covering maps $Y_{k+1} \rightarrow Y_k$ inducing transfer maps on vertex algebras $A_k = C(V_{Y_k})$ and hence a directed system

$$A_0 \hookrightarrow A_1 \hookrightarrow A_2 \hookrightarrow \cdots, \quad A_\infty := \varinjlim_k A_k \cong C(\varprojlim_k V_{Y_k}),$$

with $\varprojlim_k V_{Y_k}$ a profinite (Cantor) set carrying a residual action of $\mathrm{PGL}_2(\mathbb{Z}_p)$. Then:

- (i) the Hecke–Dirac operators $D_{p,k}$ are compatible under the transfer maps up to a controlled error, defining a class $[D_p]_\infty \in KK^0(A_\infty, A_\infty)$;
- (ii) $KK^0(A_\infty, A_\infty)$ is no longer a finite matrix ring, but a genuine inductive limit incorporating the p -adic representation-theoretic structure of $\mathrm{PGL}_2(\mathbb{Z}_p)$ -modules, in which the individual finite invariants $\mathrm{Jac}(Y_k) = \mathrm{Tors} \, \mathrm{coker}(\Delta_{p,k})$ assemble into an inverse system with growth of Iwasawa shape. Here the two kinds of tower must be separated, and the separation is dictated by our own data (Section 5), not by preference:
 - (ii-a) *Abelian deck towers.* For towers carrying a genuine $\mathbb{Z}_p$ -deck action (the voltage towers of Table 2), the limit $\varprojlim_k \mathrm{Jac}(Y_k)[p^\infty]$ is conjectured to be a finitely generated torsion $\mathbb{Z}_p[[T]]$ -module with characteristic element the cleared voltage pencil $g(T) = \det D(1+T)$; the finite-level evidence is exact — the growth law and the Newton-polygon identities $\mu_{\mathrm{meas}} = \mu(g)$, $\lambda_{\mathrm{meas}} = \lambda(g) - 1$ of Remarks 5.1 and 5.3 hold to the integer in all four computed towers — but the limit-module statement itself remains conjectural at the level of this paper, and its attribution in the literature is flagged in the caveats.
 - (ii-b) *The congruence tower.* The tower $\Gamma_0(p^k) \backslash X$ is non-normal and carries no $\mathbb{Z}_p$ -deck action, and its canonical combinatorial model, the Iwahori path tower of Remark 5.2, exhibits the exact pseudo-Iwasawa law $\nu_2 = 6 \cdot 2^k - k - 4$ with $\lambda = -1 < 0$, which is impossible for the characteristic ideal of any finitely generated torsion $\mathbb{Z}_p[[T]]$ -module. *The naive $\mathbb{Z}_p[[T]]$ -form of this conjecture, including the tentative identification of the Ihara variable u with $1+T$ proposed in an earlier draft, is therefore withdrawn as refuted in shape by its own model.* What we conjecture in its place is that the correct base object for the congruence tower is the Iwahori–Hecke algebra $\mathcal{H}(\mathrm{PGL}_2(\mathbb{Q}_p), \mathcal{I})$ of the tower, acting through the degeneracy correspondences, with the finite invariants governed by a characteristic ideal over that non-commutative base, and the observed

- k term arising as an Euler-characteristic correction — the removal of one Eisenstein rank per level — external to the module invariants proper. We do not know the correct precise formulation, and we prefer to say so rather than to state a formula the data has already falsified;
- (iii) the trace obstruction of Theorem 1.5 persists at every finite level k , so the entire tower consists of virtual, non-realizable classes; whether the *limiting* object $[D_p]_\infty$ is subject to an analogous obstruction, or whether the profinite limit process itself provides the missing room for a genuine realization, is open.

Remark 3.7. We state Conjecture 3.6 as a research programme, not a theorem, and we have proved none of its parts; part (ii-a) is the best supported, part (ii-b) is deliberately left without a formula. For part (iii) one sharp dichotomy can already be posed, and it is the intended opening theorem of a sequel: if the limit carries a faithful invariant trace τ (type II regime), the obstruction of Theorem 1.5 should persist as non-negativity of the τ -weighted second moment of any would-be correspondence, killing realizations exactly as at finite level; if the limit is traceless (type III regime, as the boundary crossed products of Paper I in fact are), the numerical obstruction has no formulation at all, and the question becomes whether absence of an obstruction can be converted into an actual construction. Part (iii) is, in our view, the most important open question raised by this paper: it asks whether the No-Go theorems of Section 1, proved level-by-level, survive or dissolve in the projective limit. We do not know the answer, and we think this — rather than any further finite-graph computation — is the correct next object of study.

4. COMPUTATIONS

Y	n	q	$\text{Tr}(C^2)$ (computed)	$-n(q-1)$ (formula)
K_4	4	2	−4	−4
K_5	5	3	−10	−10
K_6	6	4	−18	−18
$K_{3,3}$	6	2	−6	−6
Q_3 (cube)	8	2	−8	−8
prism $C_4 \times K_2$	8	2	−8	−8
Wagner V_8	8	2	−8	−8
prism $C_5 \times K_2$	10	2	−10	−10
Möbius V_{10}	10	2	−10	−10
Petersen	10	2	−10	−10
Heawood	14	2	−14	−14

TABLE 1. $\text{Tr}(C^2)$ computed directly from the $2n \times 2n$ matrix C , against the closed form of Lemma 1.1, for every graph of Paper I’s Table 1. Exact agreement in all eleven cases, confirming a fact already proved in general; the table is illustrative, not evidential.

5. A FIRST COMPUTED INSTANCE OF THE TOWER: ABELIAN $\mathbb{Z}_\ell$ -COVERS

Conjecture 3.6 concerns the congruence tower $\Gamma_0(p^k) \backslash X$, which is neither Galois nor abelian and whose explicit construction requires quaternionic lattice data we do not assemble here. But part (ii) of the conjecture makes a *shape* prediction — Iwasawa-type growth of the torsion — and that shape can be tested rigorously today in the nearest fully constructible situation: abelian

$\mathbb{Z}/\ell^k$ -covers of a fixed base graph, built from a voltage assignment $\alpha: E_Y \rightarrow \mathbb{Z}$ on the chords of a spanning tree, with derived graphs Y_k on $n\ell^k$ vertices and compatible covering maps $Y_{k+1} \rightarrow Y_k$. Every Y_k is again $(q+1)$ -regular, so the entire theory of Papers I–II applies level by level; in particular the obstruction density of Theorem 1.5 is constant along the tower, $\text{Tr}(C_k^2)/n_k = -(q-1)$ for every k , so no naïve normalization dilutes the No-Go theorem in the limit.

Two independent exact methods were used and cross-checked at every level where both run: the reduced-Laplacian determinant and Smith normal form of the derived graph directly, and the character factorization

$$\kappa(Y_k) = \frac{\kappa(Y_0)}{\ell^k} \prod_{j=1}^{\ell^k-1} \det D(\zeta_{\ell^k}^j), \quad D(t) := (q+1)I_n - A(t),$$

with $A(t)$ the voltage adjacency pencil, the product computed exactly as a resultant of integer polynomials. The two methods agreed exactly in all twelve cross-checked levels.

voltages on K_4	ℓ	μ	λ	ν	exact law, verified range
(1, 0, 0)	2	3	1	1	$\text{ord}_2 \kappa(Y_k) = 3 \cdot 2^k + k + 1, 2 \leq k \leq 9$
(1, 1, 1)	2	0	3	4	$\text{ord}_2 \kappa(Y_k) = 3k + 4, 0 \leq k \leq 9$
(1, 2, 0)	2	2	3	3	$\text{ord}_2 \kappa(Y_k) = 2^{k+1} + 3k + 3, 2 \leq k \leq 9$
(1, 0, 0)	3	0	1	0	$\text{ord}_3 \kappa(Y_k) = k, 0 \leq k \leq 6$

TABLE 2. Exact ℓ -adic valuations of $|\text{Jac}(Y_k)| = \kappa(Y_k)$ along four $\mathbb{Z}/\ell^k$ voltage towers over K_4 ($q = 2$). In each tower the Iwasawa law $\text{ord}_\ell \kappa(Y_k) = \mu\ell^k + \lambda k + \nu$ holds *exactly* for every computed level in the stated range; the parameters were fitted from two increments and the remaining six to eight levels are genuine verification, not fit.

Remark 5.1 (What the table shows, and what it does not). Three points, in decreasing order of confidence. (1) The growth law itself, for abelian ℓ -towers of graphs, is a known theorem of Vallières and of McGown–Vallières [5, 6], with closely related Iwasawa-theoretic structure in Gonet [7]; the bibliographic records are now verified and their titles corroborate the attribution, though the theorems themselves have not been checked against the papers’ texts here. Our computations *confirm* that theory in four instances; they do not discover it. (2) The Smith normal forms refine the order-growth law structurally: in the $\mu = 0$ tower (1, 1, 1) the 2-primary part of $\text{Jac}(Y_k)$ is, for $1 \leq k \leq 3$,

$$\text{Jac}(Y_k)[2^\infty] \cong \mathbb{Z}/2^k \oplus \mathbb{Z}/2^{k+2} \oplus \mathbb{Z}/2^{k+2},$$

exactly $\lambda = 3$ cyclic factors each growing by one power of ℓ per level, while in the $\mu = 3$ tower the *number* of cyclic factors itself grows (5 factors at $k = 2$, 9 at $k = 3$): structurally, λ counts persistently growing cyclic factors and $\mu > 0$ manifests as exponential proliferation of new ones. We do not know whether this structural statement in this form appears in the graph-Iwasawa literature. (3) Positive μ occurs freely here ($\mu = 3$ and $\mu = 2$ above), in genuine contrast to the cyclotomic number-field setting, where $\mu = 0$ is the Ferrero–Washington theorem; whatever the correct formulation of Conjecture 3.6(ii) for the congruence tower turns out to be, it cannot import the $\mu = 0$ expectation from number fields.

Remark 5.2 (The Iwahori path tower: exact data against the naive conjecture). The canonical combinatorial model of the congruence tower itself is constructible: take level- k vertices to be the non-backtracking paths of length k in Y_0 (level 1 is the Hashimoto digraph of A^{nb} , level $k \geq 2$

its iterated line digraphs), so that $|V(Y_k)| = n(q+1)q^{k-1}$ reproduces the index $[\Gamma(1) : \Gamma_0(p^k)] = (p+1)p^{k-1}$ exactly. For $Y_0 = K_4$, $q = 2$, the sandpile groups of these Eulerian digraphs were computed exactly for $0 \leq k \leq 6$ (up to 384 vertices). Two exact laws emerge. First, a Levine-type recursion, $|\text{sandpile}(Y_{k+1})| = 2^{n_k-1} |\text{sandpile}(Y_k)|$, verified at five consecutive levels. Second, the valuation obeys a pseudo-Iwasawa law exactly for $2 \leq k \leq 6$:

$$\nu_2(|\text{sandpile}(Y_k)|) = 6 \cdot 2^k - k - 4, \quad \text{i.e.} \quad (\mu, \lambda, \nu) = (6, -1, -4).$$

The invariant $\lambda = -1$ is *negative*, which is impossible for the characteristic ideal of a finitely generated torsion $\mathbb{Z}_p[[T]]$ -module. The canonical combinatorial model of the congruence tower therefore *refutes the naive $\mathbb{Z}_p[[T]]$ -form of part (ii) of Conjecture 3.6 as stated above*: whatever governs the congruence tower, it is not an ordinary Iwasawa module — consistently with the fact that this tower carries no $\mathbb{Z}_p$ -deck action, being non-normal. The -1 has a transparent source, the removal of one Eisenstein (trivial-direction) rank per level, an Euler-characteristic correction rather than a module invariant; the corrected form of the conjecture should be stated either over the Iwahori–Hecke algebra of the tower or as the abelian-deck statement, which the data of Table 2 and the Newton test below confirm exactly. Structurally, the Smith normal forms at $k \leq 3$ show the same three persistent growing factors $\mathbb{Z}/2^k \oplus (\mathbb{Z}/2^{k+2})^2$ as the abelian $(1, 1, 1)$ tower, plus an exponentially proliferating cloud of bounded factors (8 copies of $\mathbb{Z}/2$ at $k = 2$; 12 copies of $\mathbb{Z}/2$ and 8 of $\mathbb{Z}/4$ at $k = 3$): λ counts persistent factors, μ the proliferation rate, and the negative correction lives outside both.

Remark 5.3 (Newton-polygon confirmation of the characteristic element). In the four abelian towers of Table 2 the characteristic-element mechanism was tested directly: with $g(T)$ the cleared voltage pencil $\det D(1+T) \in \mathbb{Z}[T]$, the Newton data give, in all four cases exactly,

$$\mu_{\text{measured}} = \mu(g), \quad \lambda_{\text{measured}} = \lambda(g) - 1,$$

and the uniform offset -1 is fully accounted: $t \mapsto t^{-1}$ symmetry of the pencil forces a double zero $T^2 \mid g$ at the trivial character, contributing $+2k$ to the valuation sum, while the normalization $\kappa(Y_k) = \kappa(Y_0)\ell^{-k} \prod_{\zeta \neq 1} \det D(\zeta)$ removes k ; net $+k$, i.e. $\lambda(g) - 2 + 1$. Within the abelian regime the graph-theoretic p -adic L -function is thus confirmed to the integer.

Remark 5.4 (Relation to Conjecture 3.6). These towers are a rigorous toy, not the conjecture: the congruence tower is non-normal and its covering combinatorics are those of the full tree, not of an abelian voltage group. What the computation establishes is only this: the Iwasawa shape asserted in part (ii) is exactly right in the nearest constructible abelian analogue, down to the integer, with the characteristic-ideal mechanism (here, the resultant of the voltage pencil against $t^{\ell^k} - 1$) functioning precisely as the graph-theoretic p -adic L -function. Part (iii) — the fate of the trace obstruction in the projective limit — is untouched by any of this, and remains the paper’s central open question.

ACKNOWLEDGEMENT OF PROVENANCE AND CAVEATS

This is an unrefereed preprint. Theorem 1.2 and Theorem 3.3 are proved here in full from elementary linear algebra and require no external citation. Theorem 1.5 depends on the claim that shift equivalence forces agreement of the full nonzero spectrum of two integer matrices, which we attribute to the resolution of the Boyle–Handelman spectral conjecture by Kim, Ormes and Roush (both names corrected from an earlier draft’s “Boyer” and “Ozawa”); this attribution has not been checked against the primary literature, since the authors had no network access while writing this paper, and should be verified independently before further citation. Proposition 2.1 is an argument, not a theorem in the usual sense: it asserts the non-existence of a construction, supported by an impossibility argument (such a functor would trivialize Theorems 1.2–1.5) rather than a proof ruling out every conceivable construction. Conjecture 3.6 is entirely open; none of

its three parts is established here, and part (iii) is explicitly posed as the paper’s main unresolved question. All computations of Section 4 are exact and were checked independently of the general proof of Lemma 1.1. The tower computations of Section 5 are exact integer computations, cross-checked between two independent methods at twelve levels. Bibliography status: every external entry is now verified against its primary source or its Crossref record, with the DOI recorded in place. The three entries that had returned mutually contradictory data in two earlier out-of-band passes (Vallières, McGown–Vallières, Gonet) are resolved, and the resolution is chastening: *all three* earlier variants were wrong, in the Gonet case down to the title and the journal. Residual soft spots are noted inline: the Williams end page (Crossref records the start page only), and the Pimsner chapter year (1996 electronic versus 1997 volume). One distinction is maintained deliberately: what has been verified is the bibliographic *metadata*; the attribution of the abelian growth law to these works rests on their (now corroboratively titled) records and on recollection of their content, not on a reading of their theorems performed within this verification pass. The possibility that the structural refinement of Remark 5.1(2) is known to their authors stands as before. Two author-name errata were applied: “Boyer–Handelman” corrected to Boyle–Handelman, and “Kim–Ozawa–Roush” corrected to Kim–Ormes–Roush; the latter was the present authors’ own error, and both corrected names should themselves be confirmed at the primary-source stage.

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